

MATH2015 – Perturbation Method

Matched Asymptotic Method

In this document, the solution to the following boundary value problem is investigated:

$$\varepsilon y''(x) - 2(x^2 + 1)y'(x) + 1 = 0, \quad x \in [0, 1], \quad (1)$$

$$\text{with } y(0) = 0, \quad y(1) = 0, \quad 0 < \varepsilon \ll 1. \quad (2)$$

In Equation 1, from the fact that the highest derivative of the problem has a small number as a prefactor, one clearly identifies the problem to be singular. A convenient way to solve this problem is to exploit the appearance of small numbers by using a matched asymptotic method.

This document uses this method to analytically solve this problem with a leading order accuracy plus one correction and from there find a composite approximated solution. To do so, the investigation of the location and size ?? of the boundary value problem is realized in section 1. From there, both the outer and inner solution are respectively determined in ?? and ??. The composite solution is then obtained by matching both solutions in ??.

Following the development of the analytical solution, a numerical solution is obtained and compared to the analytical one in ?? to assess and compare the quality of both methods.

1 Identification of the location and size of the boundary layer

To solve this problem it is interesting to start by investigating the problem's boundary layer. For a second order PDE, the location of this region is determined by the sign of the prefactor of the first derivative which in this case is :

$$a(x) = -2(x + 1) \in [-4, -2]. \quad (3)$$

Because this term is negative for all possible values of $x \in [0, 1]$, the boundary layer is located about $x = 1$. **size of the BL ?**

2 Outer expansion

First, let us investigate the regular expansion away from the boundary layer i.e. away from $x = 1$. To do so, one may propose to seek for a solution using the regular Ansatz:

$$y(x) \sim y_0(x) + \varepsilon y_1(x) + \dots \quad (4)$$

Substituting Equation 4 into the initial problem Equation 1 gives :

$$\varepsilon(y_0'' + \varepsilon y_1'' + \dots) - 2(x^2 + 1)(y_0' + \varepsilon y_1' + \dots) + 1 = 0. \quad (5)$$

For the outer solution to fulfill this last equation, the lastest has to be verified for all powers of ε . Let us investigate Equation 5 for increasing powers of ε until the outer solution is obtained to the order ε i.e. after unveiling y_0 and y_1 .

Order zero

Equation 5 at a first order accuracy gives:

$$-2(x^2 + 1)y_0'(x) + 1 = 0,$$

which can be rewritten as:

$$y_0'(x) = \frac{1}{2(x^2 + 1)}.$$

Integrating this last equation gives,

$$y_0(x) = \frac{1}{2} \arctan x + C_0.$$

In section 1, we showed that the boundary layer is expected at $x = 1$. Therefore the outer solution should satisfy the left boundary condition $y_0(0) = 0$ which gives $C_0 = 0$. The solution is therefore given by:

$$y_0(x) = \frac{1}{2} \arctan x. \quad (6)$$

At $x = 1$,

$$y_0(1) = \frac{1}{2} \arctan 1 = \frac{\pi}{8} \neq 0,$$

so the outer solution does not satisfy the right boundary condition. This confirms the presence of a boundary layer near $x = 1$.

Order one

Equation 5 gives the following equation at order ε :

$$y_0''(x) - 2(x^2 + 1)y_1'(x) = 0,$$

which can be rewritten as:

$$y_1'(x) = \frac{y_0''(x)}{2(x^2 + 1)}. \quad (7)$$

Using the Equation 6, the first derivatives of the first order outer solution y_0 can be evaluated and give:

$$y_0'(x) = \frac{1}{2(1 + x^2)}, \quad y_0''(x) = -\frac{x}{(1 + x^2)^2}.$$

By injecting this result in Equation 7, one gets:

$$y_1'(x) = -\frac{x}{2(1 + x^2)^3}.$$

Integrating this last equation allow us to find:

$$y_1(x) = \frac{1}{8(1 + x^2)^2} + C_1.$$

The constant C_1 is choosen so that the outer expansion satisfies the left boundary condition order by order:

$$y_1(0) = 0.$$

Thus

$$0 = \frac{1}{8} + C_1 \implies C_1 = -\frac{1}{8}.$$

Therefore

$$y_1(x) = \frac{1}{8(1+x^2)^2} - \frac{1}{8}. \quad (8)$$

So the outer expansion through first correction is

$$y_{\text{out}}(x) = \frac{1}{2} \arctan x + \varepsilon \left(\frac{1}{8(1+x^2)^2} - \frac{1}{8} \right). \quad (9)$$

3 Inner expansion

To investigate the inner solution of our problem, one might think of using stretched coordinate of the form:

$$X = x/\delta, \quad (10)$$

which in the case of $\delta = \varepsilon$ allows us to rewrite the problem's equation as a regular equation of the stretched coordinate. Moreover, since the boundary layer is located near $x = 1$, it is interesting to use a rotated coordinate of the type :

$$X = \frac{1-x}{\varepsilon}. \quad (11)$$

The problem may then be rewritten for the new variable $Y(X)$ with

$$y(x) = Y(X).$$

The first derivatives of $y(x)$ may be expressed using $Y(X)$ as

$$\frac{dy}{dx} = -\frac{1}{\varepsilon} Y'(X), \quad \frac{d^2y}{dx^2} = \frac{1}{\varepsilon^2} Y''(X).$$

Substituting into Equation 1 and using the relation $x = 1 - \varepsilon X$ gives

$$\varepsilon \cdot \frac{1}{\varepsilon^2} Y'' - 2((1 - \varepsilon X)^2 + 1) \left(-\frac{1}{\varepsilon} Y' \right) + 1 = 0,$$

or

$$\frac{1}{\varepsilon} Y'' + \frac{2}{\varepsilon} ((1 - \varepsilon X)^2 + 1) Y' + 1 = 0.$$

Multiplying by ε ,

$$Y'' + 2((1 - \varepsilon X)^2 + 1) Y' + \varepsilon = 0.$$

This last expression is a regular reformulation of the inner boundary problem which is solvable using a simple regular Ansatz of the stretched and rotated formulation of the type:

$$Y(X) \sim Y_0(X) + \varepsilon Y_1(X) + \dots$$

Also,

$$2((1 - \varepsilon X)^2 + 1) = 2(1 - 2\varepsilon X + \varepsilon^2 X^2 + 1) = 4 - 4\varepsilon X + O(\varepsilon^2).$$

So the inner equation becomes

$$Y'' + (4 - 4\varepsilon X + \dots) Y' + \varepsilon = 0.$$

Order $O(1)$

$$Y_0'' + 4Y_0' = 0.$$

The general solution is

$$Y_0(X) = A_0 e^{-4X} + B_0.$$

Using the boundary condition inside the inner layer i.e. for $x = 1$, one gets:

$$y_0(x = 1) = Y(X = 0) = A_0 + B_0 = 0.$$

Thus, the solution may be rewritten as,

$$Y_0(X) = B_0(1 - e^{-4X}),$$

The remaining constant coefficient B_0 may be determined using a matching condition with the outer solution around $x = 1^-$. This matching condition is imposed somewhere outside the boundary layer meaning that $1 - x \gg \varepsilon$ which allows us to consider $X \rightarrow \infty$. The matching condition therefore writes:

$$\lim_{x \rightarrow 1^-} y_{out}(x) = \lim_{X \rightarrow +\infty} Y_0(X).$$

Using the first order solutions for the inner and outer regions, one gets:

$$\lim_{x \rightarrow 1} \frac{1}{2} \arctan(x) = \lim_{X \rightarrow +\infty} B_0(1 - e^{-4X}),$$

which allows to identify $B_0 = \pi/8$. Finally, one can write:

$$Y_0(X) = \frac{\pi}{8}(1 - e^{-4X}). \quad (12)$$

Order $O(\varepsilon)$

At $O(\varepsilon)$,

$$Y_1'' + 4Y_1' - 4X Y_0' + 1 = 0.$$

Since

$$Y_0'(X) = \frac{\pi}{2} e^{-4X},$$

we get

$$Y_1'' + 4Y_1' - 2\pi X e^{-4X} + 1 = 0.$$

Let

$$Z(X) = Y_1'(X).$$

Then

$$Z' + 4Z = 2\pi X e^{-4X} - 1.$$

Multiply by the integrating factor e^{4X} :

$$\frac{d}{dX}(e^{4X} Z) = 2\pi X - e^{4X}.$$

Integrating,

$$e^{4X} Z = \pi X^2 - \frac{1}{4} e^{4X} + C,$$

so

$$Z = Y_1' = \pi X^2 e^{-4X} - \frac{1}{4} + C e^{-4X}.$$

Integrating again,

$$Y_1(X) = -\frac{X}{4} + D - e^{-4X} \left(\frac{\pi}{4} X^2 + \frac{\pi}{8} X + \frac{\pi}{32} + \frac{C}{4} \right).$$

Now we match with the outer expansion near $x = 1$. First expand the outer solution in the inner variable X .

$$y_0(1 - \varepsilon X) = y_0(1) - \varepsilon X y_0'(1) + O(\varepsilon^2),$$

and

$$y_1(1 - \varepsilon X) = y_1(1) + O(\varepsilon).$$

Thus the outer expansion near $x = 1$ is

$$y_{\text{out}} \sim \frac{\pi}{8} + \varepsilon (-X y_0'(1) + y_1(1)).$$

Now

$$y_0'(1) = \frac{1}{2(1+1^2)} = \frac{1}{4},$$

and

$$y_1(1) = \frac{1}{8(1+1)^2} - \frac{1}{8} = \frac{1}{32} - \frac{1}{8} = -\frac{3}{32}.$$

Therefore the matching condition is

$$Y_1(X) \sim -\frac{X}{4} - \frac{3}{32} \quad \text{as } X \rightarrow \infty.$$

Hence

$$D = -\frac{3}{32}.$$

Next impose the boundary condition at $X = 0$:

$$Y_1(0) = 0.$$

So

$$0 = -\frac{3}{32} - \left(\frac{\pi}{32} + \frac{C}{4} \right),$$

which gives

$$\frac{C}{4} = -\frac{\pi + 3}{32}.$$

Substituting this into the formula for Y_1 , the bracket simplifies:

$$\frac{\pi}{32} + \frac{C}{4} = \frac{\pi}{32} - \frac{\pi + 3}{32} = -\frac{3}{32}.$$

Therefore

$$Y_1(X) = -\frac{X}{4} - \frac{3}{32} + e^{-4X} \left(\frac{3}{32} - \frac{\pi}{8} X - \frac{\pi}{4} X^2 \right).$$

Thus the inner expansion through first correction is

$$y_{\text{in}}(x) = \frac{\pi}{8} (1 - e^{-4(1-x)/\varepsilon}) + \varepsilon \left[-\frac{1-x}{4\varepsilon} - \frac{3}{32} + e^{-4(1-x)/\varepsilon} \left(\frac{3}{32} - \frac{\pi}{8} \frac{1-x}{\varepsilon} - \frac{\pi}{4} \frac{(1-x)^2}{\varepsilon^2} \right) \right].$$

It is cleaner to leave this in terms of $X = (1-x)/\varepsilon$:

$$Y(X) = \frac{\pi}{8} (1 - e^{-4X}) + \varepsilon \left[-\frac{X}{4} - \frac{3}{32} + e^{-4X} \left(\frac{3}{32} - \frac{\pi}{8} X - \frac{\pi}{4} X^2 \right) \right].$$

4 Matching and composite expansion

The common part (overlap) of the outer and inner expansions is obtained by taking:

$$X \rightarrow \infty \quad \text{in the inner expansion,}$$

or equivalently

$$x \rightarrow 1 \quad \text{in the outer expansion.}$$

From above,

$$Y_0(X) \sim \frac{\pi}{8}, \quad Y_1(X) \sim -\frac{X}{4} - \frac{3}{32}.$$

So the overlap is

$$y_{\text{overlap}} = \frac{\pi}{8} + \varepsilon \left(-\frac{X}{4} - \frac{3}{32} \right), \quad X = \frac{1-x}{\varepsilon}.$$

The composite expansion is

$$y_{\text{comp}} = y_{\text{out}} + y_{\text{in}} - y_{\text{overlap}}.$$

Leading-order composite solution

At leading order,

$$y_{\text{comp}}^{(0)}(x) = \frac{1}{2} \arctan x + \frac{\pi}{8} (1 - e^{-4X}) - \frac{\pi}{8},$$

hence

$$y_{\text{comp}}^{(0)}(x) = \frac{1}{2} \arctan x - \frac{\pi}{8} e^{-4(1-x)/\varepsilon}.$$

Composite solution including one correction

At $O(\varepsilon)$, the overlap subtraction removes the terms

$$-\frac{X}{4} - \frac{3}{32},$$

so the exponentially small inner remainder is simply

$$e^{-4X} \left(\frac{3}{32} - \frac{\pi}{8} X - \frac{\pi}{4} X^2 \right).$$

Therefore the uniformly valid composite approximation is

$$y_{\text{comp}}(x) = \frac{1}{2} \arctan x - \frac{\pi}{8} e^{-4(1-x)/\varepsilon} + \varepsilon \left[\frac{1}{8(1+x^2)^2} - \frac{1}{8} + e^{-4(1-x)/\varepsilon} \left(\frac{3}{32} - \frac{\pi}{8} \frac{1-x}{\varepsilon} - \frac{\pi}{4} \frac{(1-x)^2}{\varepsilon^2} \right) \right].$$

This is the matched asymptotic composite solution, accurate to leading order plus one correction.

5 Numerical results comparison

6 Use of artificial intelligence tools